

General product information



Preliminary Data Sheet

Elastollan® Softouch 50 A 10

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Characteristic:

Thermoplastic Polyester Polyurethane Elastomers with excellent mechanical properties and wear resistance, good damping characteristics and a high resilience performance.

Processable by injection moulding.

Property	Value	Unit	Test method according to
Hardness	53	Shore A	DIN ISO 7619-1 (3s)
Tensile strength	30	MPa	DIN 53504-S2
Elongation at break	900	%	DIN 53504-S2
Tear strength	45	N/mm	DIN ISO 34-1Bb
Abrasion loss	25	mm ³	DIN ISO 4649-A

Test plates are manufactured by injection moulding from pre-dried pellets (water content less than 0.02%). Test plates are aged 20 hrs at 100°C. Specimens are cut from test plates. Test conditions: 23°C± 2°C and 50% ± 6% rel. humidity.

These are general guidance data. No statement regarding specific properties. All supplies are subject to detailed specifications to be agreed-up in each individual case and containing, among others, the tolerances to be specified therein.

Delivery form and packing:

Lentil shaped pellets

PE-Bags, 25 kg net

Octabins with PE liner bags, 1000 kg net

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Drying recommendations:

Material must be dried before processing for 2-3 hours at 80-90°C in a dehumidified air dryer. Additives have to be dried with the granules. The water content of the granules should not exceed 0,02%.

Injection moulding:

When injecting the melt should be bubble and foam free, if not we recommend to adjust the drying temperature accordingly.

Following temperatures are guide values, showing the tendency of temperature profile. These may vary depending on kind of machine and mould design.

Feed [°C]	Zone1 [°C]	Zone2 [°C]	Zone3 [°C]	Zone4 [°C]	Nozzle [°C]	Melt-temp [°C]Ca.	Mould-temp[°C] Ca.
40	165 - 175	170 - 180	175 - 185	180 - 190	180 - 190	180 - 190	20 - 40

General Recommendations:

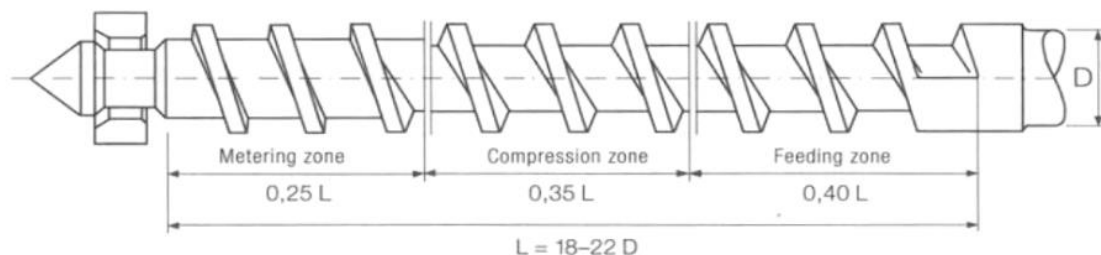
Circumferential speed (screw speed)		< 0.2 m/s 12 m/min			
Holding pressure (Melt pressure)		400 - 600 Bar			
Injection speed		relatively low			
Retention time of melt (including hot-runner)		< 10 min			
Screw speed	d _{screw} [mm]	30	45	50	60
	n _{max} [rpm]	125	80	70	60

To facilitate demoulding, mould surface with a roughness height of approx. 25-35 µm is recommended.

Ejectors should be two or three times larger than for harder thermoplastics.

Machine Design:

Injection moulding machines with single-flighted, 3-zone screws are suitable for the processing of Elastollan®. Short compression-zone screws are not suitable. The compression ratio should be around 1:2 and should not exceed 1:3. A check ring (shut-off ring) should be incorporated.



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Processing:

In cool and dry storage and in the original, undamaged and sealed containers, the products are processable for at least 6 months from delivery date. Thereafter, we do not give any warranty or guarantee regarding the processability and/or shelf life of the products. Warranties regarding buyer's rights in case of defects remain unaffected hereby.

Storage:

Elastollan® is hygroscopic, therefore storage in dry conditions and original container is recommended. Additional information about drying, processing temperatures and post-treatment are given in our product brochure "Thermoplastic Polyurethane Elastomers (TPU) Elastollan®-Processing Recommendations".

Hazard indication:

No particular hazards known. Please have a look at the Material Safety Data Sheet before handling.

Waste Disposal:

More detailed information is provided in our country-specific pamphlet and the Material Safety Data Sheet.

Important Information:

There are national and international laws and regulations to consider if it is intended to produce consumer articles (e.g. articles that necessitate food or skin contact, toys etc.) or medical objects from BASF Polyurethane Specialties (China) Company Ltd products. Where specific regulations do not exist, the current legal requirements of the European Union for consumer articles as well as medical products should be used as reference. Consultation with the BASF Polyurethane Specialties (China) Company Ltd Sales Office and our Ecology and Product Safety Department is strongly recommended.

The data contained in this document as well as advice or other support services are based on our current knowledge and experience. In view of many factors that may affect processing and application of our products, this data does not relieve processors from carrying out their own investigations and tests, particularly with regards to the suitability of the goods supplied for the processes and purposes they intend to use them for; neither does this data imply any guarantee of certain properties, or the suitability of the product for a specific purpose. Any descriptions, drawings, photographs, data, proportions, weights, measured values etc. given herein may change without prior notice and do not constitute the agreed contractual quality of the product. It is the responsibility of the recipient of our products to ensure that any proprietary rights and existing laws and legislation are observed.

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